Web Application Development

IS4102

Lecture Note 02 - Web Browsers

Presented By: 22FIS0460, 22FIS0461, 22FIS0462, 22FIS0551, 22FIS0559

INTRODUCTION

What is a Browser?

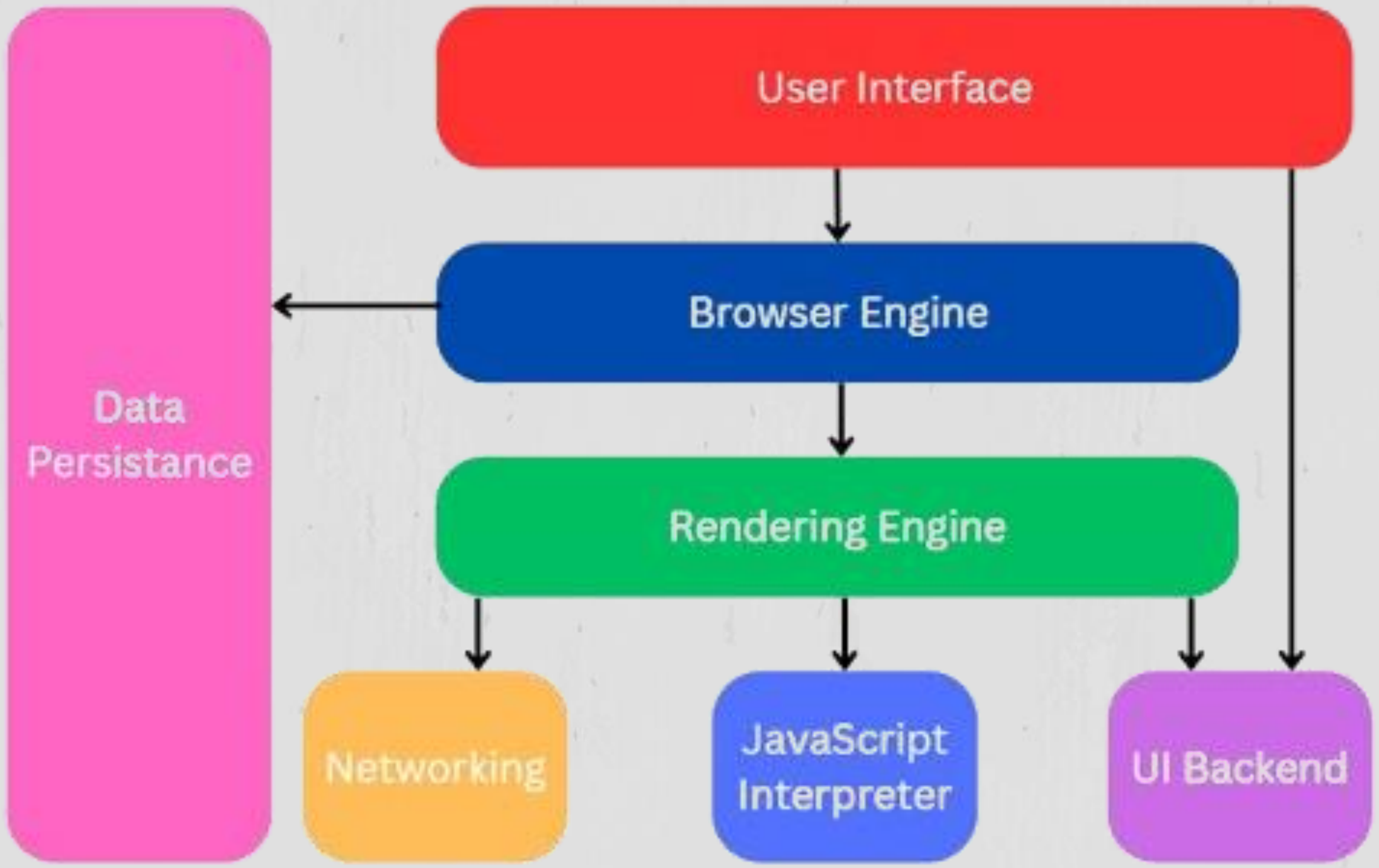
- A software application used to access, retrieve, and view content on the World Wide Web.
- It acts as an interface between the user and the internet, translating code (HTML, CSS) into interactive pages.

Some Well-Known Browsers

Safari	Firefox	Chrome	Edge	Opera
Apple	Mozilla	Google	Microsoft	Opera Software
MacOS, iOS	MacOS, MS Windows, Linux OS, Android OS	MacOS, MS Windows, Linux OS, Android OS, Chrome OS	MS Windows, MacOS, iOS, Android OS	MacOS, MS Windows, Linux OS, Android OS

BROWSER ARCHITECTURE

- The internal structure and components that work together to display web pages.
- Ensures that web pages load quickly, function smoothly, and provide a better user experience.



HOW IT WORKS

URL to Page Load

Step By Step

1. You type URL (e.g., www.google.com)
2. Browser checks **cache** (saved copy?)
3. **DNS lookup** → domain name → IP address
4. Browser sends **HTTP/HTTPS request** to server
5. Server responds with **HTML, CSS, JS files**
6. Browser begins rendering

URL → DNS → Request → Response → Render

HOW IT WORKS

Rendering

1. Parse HTML → Create **DOM tree** (structure of page)
2. Parse CSS → Create **CSSOM tree** (styles)
3. Combine → **Render tree** (what to show)
4. **Layout** → Calculate position/size of each element
5. **Paint** → Fill pixels on screen

URL → DNS → Request → Response → Render

ADVANTAGES OF WEB BROWSERS

1. Free and easy to access the web
2. Support for modern web apps (email, video calls, docs)
3. Cross-platform (Windows, Mac, Android, iOS)
4. Extensions & add-ons (ad blockers, password managers)

DISADVANTAGES OF WEB BROWSERS

1. High memory/RAM usage (Chrome is famous for this)
2. Privacy concerns – tracking cookies, data collection
3. Security vulnerabilities – zero-day exploits, malware
4. Battery drain on laptops (especially Chrome)

SECURITY FEATURES

1. **HTTPS/TLS** - Encrypts data between you and the website
2. **Sandboxing** - Each tab runs in isolation
3. **Same-Origin Policy** - Prevents one website from stealing another website's data
4. **Popup blocker** - Blocks unwanted or malicious pop-up windows automatically
5. **Phishing protection** - Warns you about fake login pages and scam sites
6. **Auto-updates** - Fixes security holes quickly without you needing to do anything
7. **Private/Incognito mode** - Doesn't save history, cookies, or form data locally

RISK AND THREATS

1. **Cross-Site Scripting (XSS)** - Hacker injects malicious script into a legit webpage
2. **Cross-Site Request Forgery (CSRF)** - Tricks your browser into doing something on another site
3. **Tracking cookies** - Companies follow you across different websites to build a profile
4. **Man-in-the-Middle** - On unsecured Wi-Fi, someone intercepts data between you and the site
5. **Drive-by downloads** - Malware downloads without you clicking anything
6. **Fake browser extensions** - Malicious add-ons that steal passwords or browsing history

QUICK COMPARISON

Browser	RAM Usage	Privacy	Best For
Chrome	High	Moderate	Speed & Extensions
Firefox	Medium	Strong	Privacy
Safari	Low	Strong	Mac battery life
Edge	Medium	Moderate	Windows Integration

GROUP MEMBERS

- 22FIS0460
- 22FIS0461
- 22FIS0462
- 22FIS0551
- 22FIS0559